

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – C Code Output Prediction

Course Code:	CSA0309	Course Title:	Data Structures
CO Assessed:	CO1 – Imitation (BL3, BL4)	Assessment:	Assessment Tool 1 – C Code Output Prediction
Weightage:	35% of CIA	Total Marks	100
CO1 Statement:	Apply suitable data structures and ADTs for efficient problem solving by analyzing time and space complexity.		
Student Name:	GAUTHAM G	Reg. No.:	192524371
Section / Batch:	SLOT -B	Date:	08/07/2026

Code of Conduct

I am GAUTHAM G 192524371 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: GAUTHAM G

Instructions

- This test contains 20 Debugging & Optimisation Task questions. Total: 100 Marks (20*5).
- When a student answer using Scenario-Based, in evaluation the answer should be with following five requirements: Identify the problem type and constraints, Select appropriate data structure, Design the Solution, Optimization and efficiency, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Linked data structure	Basic data structure	1 - 2	20
Primitive and Non-Primitive Data Structure - Linear and Non-Linear Data Structure	Classifications of Linear data structure	3 - 4	20
Search Data Structure	Linear Search and Binary Search	5 - 6	20
Recursion, Performance analysis	Recursion	7 - 8	20
Array Data Structures	Implementations Using Array data structure	9 - 10	20
GRAND TOTAL:			100 Marks

CO2_AT2_Debugging & Optimization Task (Set 2) OPEN

Debugging & Optimization Task

 0/10 answered

Attempt →

CO2_AT2_Debugging & Optimization Task (Set 1) OPEN

Debugging & Optimization Task

 0/10 answered

Attempt →

CO1_AT1_C CODE OUTPUT PREDICTION OPEN ✓ Completed

C CODE OUTPUT PREDICTION

 10/10 answered

Review →

CO2_AT2_Debugging & Optimization Task (Set 2) OPEN

Debugging & Optimization Task

 0/10 answered

Attempt →

CO2_AT2_Debugging & Optimization Task (Set 1) OPEN

Debugging & Optimization Task

 0/10 answered

Attempt →

CO1_AT1_C CODE OUTPUT PREDICTION OPEN ✓ Completed

C CODE OUTPUT PREDICTION

 10/10 answered

Review →

QUESTIONS

Q1 ✓
C Code - Pointer with Array
10 marks

Q2 ✓
C Code - Linked List Node
10 marks

Q3 ✓
C Code - Binary Search
Complexity
10 marks

Q4 ✓
C Code - Pointer Increment
10 marks

Q5 ✓
C Code - Array Address
10 marks

Q6 ✓
C Code - Linked List Traversal
10 marks

Q5 C Code - Array Address

10 marks

```
int arr[]={5,10,15};
printf("%d",*(arr+1));
```

Your Answer

10

Save Answer

QUESTIONS

Q1 ✓
C Code - Pointer with Array
10 marks

Q2 ✓
C Code - Linked List Node
10 marks

Q3 ✓
C Code - Binary Search
Complexity
10 marks

Q4 ✓
C Code - Pointer Increment
10 marks

Q5 ✓
C Code - Array Address
10 marks

Q6 ✓
C Code - Linked List Traversal
10 marks

Q8 C Code - Recursion (Sum)

10 marks

```
#include <stdio.h>
int sum(int n){
    if(n==0)
        return 0;
    return n + sum(n-1);
}
int main(){
    printf("%d", sum(5));
}
```

Your Answer

15